How Do Engineers See the Invisible?

Logs, metrics and traces

SCAN / LEARN

THE MAIN IDEA

Observability means learning what a system does from its signals. Logs record events, metrics show numbers over time, and traces follow a request through services.

FOLLOW THE SYSTEM

01

EVENT LOG

A message says what happened and when.

›

02

METRIC

A chart shows errors or response times.

›

03

TRACE

A request gets tracked across services.

›

04

DIAGNOSIS

Signals together point to a likely cause.

EVENTS + NUMBERS + PATHS → DIAGNOSIS

COMMON MISCONCEPTION

One dashboard graph tells the entire story.

MORE ACCURATE

A graph can reveal symptoms; traces and logs help find causes.

THREE QUESTIONS TO KEEP

PROBLEM: How can we investigate a hidden slowdown?

FAILURE: Without signals, incidents are guesswork.

IN REAL LIFE: Site reliability and AI serving.

SOLO DISCOVERY

Invent a slow page. What log, metric and trace would help?